Pamella Carneiro

Physical Chemist · Quality Control · Laboratory Quality Systems (ISO/IEC 17025)

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PROFILE

Analytical chemist with **3 years as Technical Manager of a physicochemical and microbiological testing laboratory**, responsible for quality review of analytical results, metrological traceability, internal audits, proficiency testing and LIMS-based data management under **ISO 9001** and **ISO/IEC 17025**. Currently a PhD candidate in Physical Chemistry (UNICAMP) and Guest Researcher at TU Delft, studying protein stability and formulation of vaccine antigens, as well as solvents of industrial and pharmaceutical interest. Combines a quality-driven laboratory mindset with strong data analysis and programming skills. Seeking **QC / analytical chemistry roles in the pharmaceutical and biotech industry** in the Netherlands.

CORE COMPETENCIES

Quality systems: ISO/IEC 17025 · ISO 9001 · quality review of analytical results · conformity assessment · metrological traceability · internal audits · proficiency testing (interlaboratory comparison) · quality procedures and analytical records · regulatory compliance

Analytical techniques: UV-Vis spectrophotometry · titrimetry (alkalinity, hardness, chlorine) · pH, conductivity, turbidity, solids, COD/BOD · metals analysis · automated discrete analyzer (Gallery) · microbiological testing (total coliforms, *E. coli*) · sampling protocols

Data & informatics: LIMS (results documentation and traceability) · data review and trending · Python · Julia · Linux · Git · reproducible, automated data-analysis workflows

Scientific background: Protein structure and stability · antigen–antibody interactions · formulation and solvation (ionic liquids, adjuvants) · molecular modelling (GROMACS)

Languages: Portuguese (native) · English (professional working proficiency)

PROFESSIONAL EXPERIENCE

Technical Manager & Laboratory Analyst

Apr 2018 – Apr 2021

LAS Sanches Laboratório · Belém, Brazil · Water and industrial wastewater testing

- Held technical responsibility for all laboratory activities; led the analysis team and coordinated laboratory and field sampling workflows.
- Performed quality review of analytical results, verifying conformity with methods, specifications and regulatory limits before reporting.
- Managed quality procedures, analytical records and metrological traceability; maintained ISO 9001 compliance and led the transition toward ISO/IEC 17025.
- Conducted internal audits and participated in proficiency testing schemes to demonstrate the reliability of results.
- Used LIMS for results documentation, data traceability and results control.
- Performed physicochemical, instrumental and microbiological analyses of groundwater, surface, drinking, recreational water and effluents.
- Ensured compliance with Brazilian regulations: CONAMA 357/2005, 396/2008, 274/2000 and 430/2011; Ministry of Health Consolidation Ordinance No. 5 (drinking water); ABNT NBR 10818.

RESEARCH EXPERIENCE

Guest Researcher

Mar 2026 – present

Delft University of Technology (TU Delft) · Delft, The Netherlands

- Calculate transport properties (self and collective diffusivities, viscosity) of industrially relevant solvents from molecular dynamics (Maxwell–Stefan, Green–Kubo).

PhD Candidate, Physical Chemistry

Feb 2024 – present

University of Campinas (UNICAMP) · Brazil

- Investigate how ionic liquids solvate and stabilise antigen proteins as candidate vaccine adjuvants (formulation stability), using Kirkwood–Buff theory.
- Built automated, reproducible GROMACS + Julia workflows, applying QC principles (traceability, validation, documentation) to simulation data.

MSc in Physical Chemistry Jan 2021 – Oct 2023

University of Campinas (UNICAMP) · Brazil

- Studied the antigen–antibody recognition mechanism of the malaria vaccine candidate PfAMA1 with a monoclonal antibody (published in *J. Phys. Chem. B*).
- Analysed how mutations shift the balance between order and disorder in protein regions.

Undergraduate Scientific Initiation, Molecular Modeling Laboratory Jan 2015 – Mar 2017

Federal University of Pará (UFPA) · Belém, Brazil

- Structure–activity studies of bioactive molecules (3D-QSAR, HQSAR, molecular docking).

EDUCATION

PhD in Physical Chemistry (in progress) — University of Campinas (UNICAMP), Brazil 2024 – present

MSc in Physical Chemistry — University of Campinas (UNICAMP), Brazil 2021 – 2023

BSc in Industrial Chemistry — Federal University of Pará (UFPA), Brazil 2014 – 2018

TEACHING & KNOWLEDGE SHARING

- Teaching assistant, Experimental General Chemistry — guided students through laboratory analyses.
- Teaching assistant, Quantum Chemistry (undergraduate course).
- Author of an open-source molecular dynamics tutorial for newcomers (github.com/pamellaccar).

PUBLICATIONS

- **P. C. C. da Silva**, L. Martínez. Extended Conformational Selection in the Antigen–Antibody Interaction of the PfAMA1 Protein. *J. Phys. Chem. B*, 2024. doi:10.1021/acs.jpcb.4c03734
- **P. C. C. da Silva**, L. Martínez. Intrinsically Disordered Malaria Antigens: An Overview of Structures, Dynamics and Molecular Simulation Opportunities. *J. Braz. Chem. Soc.* 35(10), 2024. doi:10.21577/0103-5053.20240102

AWARDS

- Highlight in Scientific and Technological Initiation, UFPA (2016).
- Scientific Initiation Summer Scholarship, Artificial Intelligence Laboratory, EACH-USP (2016).
- 3rd Place, Best Poster Award, Meeting of Chemistry Professionals of the Amazon (2016).